Diffusion Models in Finance: A Survey

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ABSTRACT

Diffusion and score-based generative models have, within a few years, spread across nearly every quantitative-finance task: synthetic data generation, limit- order-book (LOB) simulation, risk and scenario generation, derivative pricing and hedging, and portfolio construction. Their appeal is structural—stable likelihood-based training, strong mode coverage, native conditioning, and a stochastic-differential-equation (SDE) formulation that sits naturally beside the Itô calculus and stochastic control finance already uses. We present, to our knowledge, the first survey dedicated to diffusion models *in finance*. We organize a fast-growing and fragmented literature along two axes—methodological family and financial task—consolidate how diffusion's conditioning and SDE structure are exploited for finance-specific goals (risk-neutral generation, stylized-fact fidelity, controllable stress, microstructure realism), and critically review evaluation, where heterogeneous protocols, leakage, and in-distribution testing currently limit comparability. We close with a research agenda: tail fidelity, non-stationarity guarantees, hard economic constraints, efficiency for real-time use, privacy and governance, and standardized benchmarks.

KEYWORDS

diffusion models, score-based generative models, synthetic data, market simulation, limit order book, risk management, derivative pricing, portfolio optimization, stress testing

1 INTRODUCTION

Generative modeling is now central to quantitative finance, for structural rather than fashionable reasons. Financial data are *scarce* in the regimes that matter most—crises, regime transitions, and rare events are by definition under-represented in any historical sample. They are *sensitive*, since client-level and proprietary records cannot be shared freely. And they are *non-stationary*, drifting with policy, liquidity, and market structure. Synthetic data that faithfully reproduces the statistical signature of markets relaxes all three constraints at once: it augments thin datasets, substitutes for privacy-restricted records, and supplies controllable scenarios for backtesting and stress testing [30, 59].

For roughly a decade this need was met primarily by generative adversarial networks (GANs) and variational autoencoders (VAEs). Quant GANs [78] showed that temporal convolutional GANs reproduce volatility clustering and fat tails; signature-based Sig-Wasserstein GANs [52] and structured-noise/MMD models [51] improved stability and path realism. Yet GANs remain hard to train, prone to mode collapse, and awkward to condition, while VAEs oversmooth the heavy tails that dominate financial risk.

Why diffusion, why now. Denoising diffusion probabilistic models (DDPMs) [28] and score-based models [66], unified under a stochastic differential equation (SDE) view [67], offer a different trade-off: stable likelihood-based training, strong mode coverage, and native, flexible *conditioning* through guidance [19, 29]. They also sit beside the objects quantitative finance already uses. A diffusion model *is* an SDE with a learned drift; sampling is the numerical solution of a reverse-time SDE or its probability-flow ODE; and training has a control-theoretic reading through a Hamilton–Jacobi–Bellman (HJB) equation on the log-density [8]. For a field built on Itô calculus, Monte Carlo, and stochastic control, this is a comfortable fit—and the reason diffusion methods have, since roughly 2023, spread across nearly every quantitative-finance task.

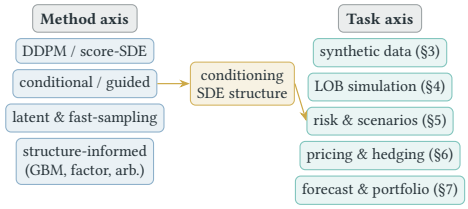


Figure 1: Two-axis taxonomy. Each surveyed work pairs a methodological family with one or more financial tasks; conditioning and SDE structure are the threads that recur across both axes.

Contributions. (i) A two-axis taxonomy (method × financial task, Figure 1) and coverage map (Figure 2) that organize the literature. (ii) A consolidated account of how diffusion's conditioning and SDE structure serve finance-specific goals—risk-neutral generation, stylized-fact fidelity, controllable stress, and microstructure realism. (iii) A critical review of evaluation and its pitfalls. (iv) A research agenda. Prior reviews are GAN/VAE-centric or target generic time series [80] and anomaly detection [49]; broad GenAI synthetic-data surveys [43] and tabular surveys [12] are not finance-specific. To our knowledge this is the first survey dedicated to diffusion *in finance*. Section 2 reviews fundamentals; Sections 3–7 survey applications; Section 8 covers emerging and cross-domain uses; Section 9 synthesizes cross-cutting themes; Section 10 treats evaluation; and Section 11 discusses open problems.

Scope. We focus on diffusion and score-based models—DDPMs, continuous score-SDEs, and their conditional, latent, fast-sampling, and structure-informed variants—applied to financial problems. We also include the *flow-matching*, *rectified-flow*, and *stochastic-interpolant* generalization (§2), of which diffusion is the Gaussian special case, but treat classical normalizing flows as a related baseline. We include the most relevant non-diffusion baselines (notably GANs and VAEs) for context but do not survey them exhaustively, and we treat cross-domain backbones only insofar as they are adapted to finance. Within each financial task we organize work by method. We prioritize peer-reviewed and widely cited results, while noting that much of this literature is recent and preprint-stage, so quantitative claims should be read as indicative.

2 BACKGROUND

This section gives the minimum machinery to read the survey and, more importantly, draws the bridge to financial mathematics.

Forward and reverse processes. A diffusion model defines a *forward* process that corrupts data $x_0 \sim p_{\text{data}}$ into noise and learns a *reverse* process that reconstructs data from noise. In DDPM [28], the forward chain adds Gaussian noise, $q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I)$, with the closed form $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon$. A network $\epsilon_\theta(x_t, t)$ is trained with the denoising objective $\mathbb{E}_{x_0,\epsilon,t} \|\epsilon - \epsilon_\theta(x_t, t)\|^2$, an idea rooted in non-equilibrium thermodynamics [63]. Score-based models [66] instead learn the score $\nabla_x \log p_t(x)$ by denoising score matching across noise scales; predicting noise and predicting the score are equivalent up to scaling. The simple objective is a reweighting of the variational bound on $\log p_\theta(x_0)$, so training is likelihood-based and stable—a sharp contrast with the adversarial min–max of GANs, and a practical reason diffusion has displaced them in finance, where training instability and mode collapse directly translate into missing market regimes. Conditioning on covariates c (a regime label, a volatility target, a history) is handled by learning $\epsilon_\theta(x_t, t, c)$ and, at sampling time, combining conditional and unconditional predictions via classifier-free guidance [29], $\tilde{\epsilon} = \epsilon_\theta(x_t, t, \varnothing) + w\,[\epsilon_\theta(x_t, t, c) - \epsilon_\theta(x_t, t, \varnothing)]$, with guidance weight w trading fidelity to c against diversity—the knob that makes "generate a crash of intensity c" operational.

The SDE unification. Song et al. [67] unify both under an Itô SDE $dx = f(x,t)\,dt + g(t)\,dW_t$ whose time reversal, $dx = [f(x,t) - g(t)^2 \nabla_x \log p_t(x)]\,dt + g(t)\,d\bar{W}_t$, is itself driven by the score, with a deterministic probability-flow ODE of identical marginals. *This is the heart of the finance connection*: asset-price models are SDEs, so a diffusion generator is a learned, data-driven SDE, and sampling is the reverse-time simulation quants already perform—now with a neural drift. The correspondence is operational, not merely aesthetic: it tells us where financial structure can be injected (the drift f, the diffusion g, the noise schedule, or the prior), how to reason about a change of measure (a drift shift, hence a score shift; §6), and why control-theoretic tools apply—the training objective satisfies an HJB equation on the evolving log-density [8], connecting generation to the optimal-investment and hedging problems of continuous-time finance. The probability-flow ODE additionally yields exact likelihoods, useful for anomaly scoring and model comparison.

Discrete vs. continuous, and which to use. The discrete DDPM and the continuous score-SDE are two views of one object, but the choice has practical consequences in finance. The discrete formulation is convenient for fixed-length tabular records and short return windows [39, 59]; the continuous SDE view is natural when the target is itself a process—asset paths, yield-curve dynamics, or order flow—because it exposes the drift and diffusion terms where economic structure (mean reversion, jumps, the risk-neutral shift) is most naturally imposed [23, 37, 72]. Few-step distillations and latent variants apply to either and are increasingly used when sampling cost is binding. In practice, financial models mix these ingredients freely: a latent-space score model with a transformer backbone, classifier-free conditioning, and a finance-specific prior is a common composite.

From recipe to practice. Building a financial diffusion model thus involves a small set of recurring choices: the *backbone* (Transformer for long-range and cross-asset dependence, convolutional/U-Net for local structure, state-space for very long sequences [4]); the *representation* (raw series, latent code, image, or byte stream); the *conditioning signal* and how it is injected (concatenation, cross-attention, or classifier-free guidance); the *noise schedule and parameterization*; and any *structural prior* encoding economic knowledge. The same few decisions recur across tasks, which is why methods transfer readily between, say, LOB simulation and scenario generation once the modality is matched. This modularity is a strength—progress in generic diffusion propagates quickly to finance—but it also means careless transfer can import assumptions (Gaussianity, stationarity, exchangeability) that financial data violate.

Sampling and acceleration. Ancestral sampling needs hundreds to thousands of network evaluations, prohibitive for high-frequency or large-scale risk use. Refinements add learned variances and improved schedules [53]; DDIM [64], EDM [36], consistency models [65], and flow matching [47] push toward few-step generation. Sampling cost recurs as a binding constraint below.

Flows and interpolants: the broader family. Flow matching is also the entry point to a *generalization* of diffusion that we treat as in scope. Rectified flow [50] learns a neural ODE transporting noise to data along near-straight paths by least-squares velocity regression, and stochastic interpolants [3] give a unifying framework in which a tunable interpolation between any two densities yields both a probability-flow ODE and a family of SDEs, with classical diffusion recovered as the Gaussian special case. For finance this matters because arbitrary base distributions and interpolation schedules let priors be built into the *path*, and the ODE form supports few-step sampling where latency binds. We therefore include flow-matching/rectified-flow/interpolant methods alongside DDPM and score-SDE, while treating *classical* normalizing flows (invertible exact-likelihood maps) as a related baseline rather than as diffusion.

Conditioning and backbones. Finance rarely wants unconditional samples; it wants samples *given* a regime, a volatility target, or a history. Classifier [19] and classifier-free [29] guidance steer generation; latent diffusion [57] improves efficiency. Cross-domain backbones recur as finance models and baselines: TimeGrad [56], CSDI [71], SSSD [4], Diffusion-TS [81], and TSDiff [38] for sequences; TabDDPM [39] and TabSyn [83] for tables; DiT [55] for transformer backbones; and Diffuser [34]/Decision Diffuser [2] (surveyed by Zhu et al. [87]) for decision-making-as-generation. The flow/interpolant family supplies its own sequence backbones—FlowTS [31] and TimeFlow [27] for generation, and interpolant-based probabilistic forecasting [14, 15]—and RecTable [20] for tables.

Why finance is hard. Generic successes do not transfer automatically. Returns exhibit heavy (power-law) tails, volatility clustering, the leverage effect, and weak return autocorrelation but strong autocorrelation of absolute returns; the distribution drifts; the signal-to-noise ratio is low; LOB data are discrete, bursty, and rule-governed; and pricing requires no-arbitrage and the correct change of measure. These tensions motivate *structure-informed* designs—GBM-based forward processes [37], factor structure [13],

arbitrage penalties [35], and risk-neutral drift corrections—that recur throughout.

3 SYNTHETIC DATA AND TIME-SERIES GENERATION

The largest cluster targets data that passes for real. Two design philosophies recur. The first *reshapes* data so off-the-shelf diffusion applies: Takahashi and Mizuno [69] apply a wavelet transform mapping price, volume, and spread into image-like representations, train a DDPM, and invert it, recovering fat tails and clustering. The second *bakes in* financial structure: Kim et al. [37] replace Gaussian corruption with a geometric-Brownian-motion forward process, improving heavy-tail and clustering fidelity. The lesson is that inductive bias—priors in the noise process or representation—outperforms architecture tuning alone. The fidelity bar is unusually demanding: a financial generator must reproduce a battery of *stylized facts* simultaneously—fat (power-law) tails, volatility clustering, the leverage effect, weak return autocorrelation alongside strong autocorrelation of absolute returns, and gain/loss asymmetry—none of which a low reconstruction error guarantees. Score networks for these tasks typically use Transformer, U-Net, or convolutional backbones with explicit temporal and positional encodings and diffusion-step embeddings to capture non-stationary temporal dependence; the multivariate case additionally demands faithful cross-sectional dependence, which motivates the factor- and correlation-structured designs that recur in §5–7.

The data modalities span a wide range, and the appropriate machinery differs by modality. Low-frequency univariate and panel returns are the most studied setting and the natural home of stylized-fact evaluation [13, 69]. High-frequency and microstructure data (§4) are discrete and event-driven, demanding event- or byte-level models [45, 86]. Mixed-type tabular records—credit, fraud, fund holdings—require combined continuous/categorical diffusion [39, 59]. And structured objects such as implied-volatility surfaces or yield curves are best modeled with the relevant no-arbitrage constraints built in [21, 35]. A practical implication is that "diffusion for finance" is not one method but a family, selected and specialized per modality; the cross-domain backbones of §2 supply the starting points.

High-dimensional and conditional generation. Portfolio and risk applications need realistic *dependence*. Chen et al. [13] embed a latent factor structure into diffusion, generating high-dimensional returns that respect a factor covariance, with theoretical guarantees and small-sample gains. Conditioning turns a generator into a scenario engine: CoFinDiff [70] controls trend and realized volatility with lower control error and more diversity than GAN/VAE baselines, and downstream hedging gains. Fidelity is ultimately judged by use: Fiaingen [58] reports train-on-synthetic/test-on-real near parity with real data, and comparative studies position diffusion against TimeGAN/VAE as a privacy-preserving substitute [30]; general style-guided TS diffusion [68] supplies further backbone machinery.

Tabular data, fraud, and privacy. Much financial data is mixed-type tabular. TabDDPM [39], latent-space TabSyn [83], and its rectified-flow analogue RecTable [20] define the general state of the art; FinDiff [59], presented at ICAIF, specializes it to financial tables (credit, expenditure, fund composition) with strong fidelity, utility, and privacy versus TVAE/CTGAN/TabDDPM. Two extensions address the field's defining difficulty—class imbalance: Imb-FinDiff [61] adds a joint denoising-plus-label objective for minority (fraud) synthesis, and EntTabDiff [48] conditions on *entities* to improve fidelity and reduce entity-level leakage. For fraud, EmDT [40] trains per-cluster diffusion transformers and latent flow-diffusion [33] targets minority oversampling. Privacy is moving from proxies to formal guarantees: DP-FinDiff [62] and federated DP-FedTabDiff [60] train under differential privacy, and Zuo et al. [88] benchmark the privacy–utility frontier. Actuarial use is adjacent (survival-data diffusion [10]). The non-diffusion frontier (adversarial synthetic data [79]) remains the comparison point.

Relation to GANs, VAEs, and flows. It is worth stating the comparison plainly, since these baselines recur throughout. GANs—Quant GANs [78], Sig-Wasserstein GANs [52], and signature-MMD models [51]—can reproduce stylized facts and were the first deep generators to do so convincingly, but their adversarial training is unstable and prone to mode collapse, and conditioning them on rich controls is awkward. VAEs train stably and give latent structure but tend to oversmooth, blunting exactly the heavy tails finance cares about. Across the surveyed studies the reported pattern is consistent: diffusion matches or exceeds these baselines on stylized-fact and downstream metrics while conditioning more naturally, at the cost of slower sampling. None of this is a clean dominance claim—cross-paper comparison is confounded by inconsistent data and metrics (§10)—but it explains the field's rapid migration toward diffusion.

4 MARKET MICROSTRUCTURE AND LOB SIMULATION

Simulating the LOB and its order flow is harder than low-frequency generation: the data are high-frequency, discrete, bursty, and rule-governed, and a useful simulator must be *responsive*—reacting plausibly to an agent's own orders, so it can serve backtesting, market-impact studies, and agent-based experiments.

Order flow and representations. TRADES [9] generates order flow with a transformer diffusion conditioned on market state, reporting predictive-score gains; since the book reconstructs from the flow, realism propagates. "Painting the Market" [6] renders the LOB as an image and applies inpainting, generating long horizons in parallel and avoiding the error accumulation of autoregressive simulators, scoring well on LOB-Bench. ByteGen [45] forgoes tokenization, generating events directly in byte space with a hybrid Mamba–Transformer.

Volume, counterfactuals, and control. Beyond price levels, the volume profile governs liquidity and impact: DiffVolume [76] generates volume snapshots with *counterfactual* control, and DiffLOB [77] produces regime-conditioned counterfactual LOB trajectories—supporting microstructure-level what-if stress. DigMA [32] pairs a conditional diffusion of the market state with a financially informed meta-agent for controllable, responsive order generation. Event-stream models (LOBDIF [86]) predict the joint timing and type of events with skip-step sampling for latency. Diffusion also serves upstream as a market-series *denoiser* [75], and lower-frequency

dynamics simulation yields portfolio-ready covariances [42]. Interpretable evaluation uses optimal transport to test power-law reproduction [26].

The defining requirement that separates microstructure simulation from mere sequence generation is *responsiveness*: under a strategy backtest or market-impact study the simulated book must react plausibly to the evaluated agent's own orders, not merely replay history. Conditioning the generator on the evolving market state—and, in DigMA [32], on an explicit agent—is the route to this, and counterfactual generation [76, 77] is its controllable form. Two constraints temper progress: latency (thousand-step sampling is infeasible at high frequency, motivating the skip-step scheme of LOBDIF [86] and few-step samplers generally) and the discrete, rule-governed nature of order events, which byte- and event-level models [45, 86] address more directly than continuous-image framings.

This work sits in a longer arc of market simulation that moved from hand-built agent-based and stochastic point-process models, through GAN-based generators, to the current diffusion frontier, which packages stable training, broad distributional coverage, and flexible conditioning. Diffusion simulators are increasingly positioned as substrates for *agent-based* experiments [9, 32]: a calibrated, responsive generator lets researchers inject hypothetical agents or shocks and observe plausible market reactions, which is precisely what counterfactual and regime-conditioned designs [76, 77] enable. The open evaluation question—how to certify that a simulator's *reaction* to an unseen agent is realistic, not merely that its unconditional samples match history—remains the key obstacle to using these simulators for high-stakes strategy validation.

5 RISK, SCENARIO GENERATION, AND STRESS TESTING

Controllability pays off most in risk, where the goal is to populate *tails* and *what-ifs* absent from history. Guidance [19, 29] yields samples *given* a crash intensity or regime; the same heavy-tail fidelity that aids generation makes the resulting Value-at-Risk and Expected-Shortfall estimates credible—a generator that thins tails understates risk. Methodologically, a trained generator is used as a Monte Carlo engine: risk measures (VaR, ES/CVaR) and their sensitivities are estimated from sampled scenarios, optionally conditioned on a stressed regime, so the quality of the tail directly determines the quality of the capital number. This makes the stylized-fact and tail-fidelity evaluation of §10 not a cosmetic check but a precondition for trustworthy risk use, and it is why several recent methods add explicitly tail-weighted training objectives or regime conditioning [5, 74] rather than relying on a generator tuned for average-case realism. Scenario generation is most useful wired into a decision: diffusion-augmented RL [17] trains robust policies against generated stress, and MARCD [5] couples a regime-aligned diffusion generator (a hidden Markov model) to a CVaR-constrained allocator with a tail-weighted objective, reporting large drawdown reductions. Structure-aware generation applies stress coherently: factor-structured diffusion [13] shocks factors while preserving dependence; compressed-sensing-accelerated diffusion in a low-dimensional macro space [25] reproduces left-tail quantiles for backtesting; diffusion scenario trees [82] feed multistage stochastic optimization; and neural Lévy jump-diffusion [74] targets jump-dominated tails. Many techniques transfer from energy scenario generation [44, 73]. Regulated stress testing could use richer, internally consistent scenario sets, but a scenario generator used for capital must itself be validated and auditable (Section 11)—in our view the main barrier between these results and regulated deployment.

6 PRICING, HEDGING, AND CALIBRATION

Pricing imposes the hardest economic constraints: no-arbitrage and the change of measure from the physical ($\mathbb{P}$) to the risk-neutral ($\mathbb{Q}$) world. A change of measure shifts the SDE drift [67], which in score/ϵ terms is an additive correction to the reverse dynamics; Tiwari [72] realize exactly this as a score-function shift enforcing the martingale condition, making the measure change explicit rather than approximate (cf. the built-in risk-neutral transition of Quant GANs [78]). Zhao and Shen [85] price exotic and path-dependent options (European, arithmetic Asian) with conditional diffusion, honestly reporting degraded accuracy on *extreme* scenarios—the crux for risk-sensitive products. Jin and Agarwal [35] generate arbitrage-aware implied-volatility surfaces one day ahead by penalizing arbitrage violations in training, beating GAN-based surface models.

Fixed income and foundations. Diffusion has reached rates: Fukunishi et al. [21] generate arbitrage-aware yield-curve term structures with macroeconomic conditioning, and Zhang et al. [84] synthesize language-conditioned Treasury-futures series. More foundationally, Gao et al. [23] generate SDE solution paths *without* known drift and diffusion coefficients—the calibration setting when the model is unknown—and the HJB view [8] links diffusion to the optimal-investment and hedging equations. Deep hedging [11] is the downstream decision these generators serve: better training distributions yield better learned hedges, as CoFinDiff [70] demonstrates. Neural PDE solvers for (rough) volatility pricing [54] are a complementary, non-generative route. Calibration—fitting a model to observed market prices—reappears throughout: generating arbitrage-consistent surfaces [35] or term structures [21] is calibration in distributional form, and learning unknown SDE coefficients from data [23] is calibration when the model class itself is uncertain. Hedging connects pricing to decisions: with a faithful $\mathbb{Q}$-generator, sensitivities (Greeks) and hedge ratios can be estimated by simulation, and deep-hedging policies [11] can be trained on diffusion-generated paths that include frictions and jumps. The unresolved frontier is enforcing *exact* no-arbitrage and the martingale property by construction—via constrained sampling, projection, or hard-constrained guidance—rather than penalizing their violation, while preserving tail accuracy.

7 FORECASTING AND PORTFOLIO CONSTRUCTION

Diffusion models are *probabilistic* forecasters—they produce predictive distributions, which is what portfolio construction and risk budgeting consume. TimeGrad [56] is the standard baseline; in finance, Diffolio [16] adds hierarchical (asset- and market-level)

Table 1: Representative diffusion-in-finance works by task. "C" marks explicit conditioning/controllability.

Work	Task	C
Takahashi–Mizuno [69]	Synthetic gen	
Diffusion Factor [13]	Hi-dim returns	
CoFinDiff [70]	Controllable gen	✓
FinDiff [59]	Tabular	✓
Imb-FinDiff [61]	Tabular (imbalance)	✓
DP-FinDiff [62]	Private tabular	✓
TRADES [9]	LOB sim	✓
Painting [6]	LOB sim	✓
DiffVolume [76]	LOB volume	✓
DiffLOB [77]	LOB counterfactual	✓
LOBDIF [86]	LOB event stream	
MARCD [5]	Stress + CVaR	✓
Risk-neutral [72]	Pricing	✓
IV surface [35]	Pricing	✓
Term structure [21]	Rates	✓
Diffolio [16]	Forecast/portfolio	✓
Factor-cond. [22]	Portfolio	✓
Dyn. portfolio [1]	Portfolio	
FlowHFT [46]	Trading (flow)	✓

attention and a correlation-guided regularizer, so the predictive distribution respects cross-sectional dependence, improving both accuracy and downstream Sharpe and certainty-equivalent. Allocation follows two routes. *Predict-then-optimize*: factor-conditioned diffusion learns the cross-sectional return distribution and optimizes mean-variance/mean-CVaR portfolios [22], with a bias–variance analysis of the conditioning-factor count [7]. *Direct* policy learning: model-free dynamic selection beats Markowitz [1], building on decision-as-generation [2, 34]; the flow-matching form of this idea, FlowHFT [46], learns an imitation-learning trading policy for high-frequency trading that adapts across market regimes. Relational and augmentation variants include DiffSTOCK [18] (conditioning on indicators and inter-stock relations) and DiffsFormer [24] (factor augmentation); chart-image forecasting [41] illustrates the role of representation. A third, indirect route is *augmentation*: synthetic paths enlarge thin training sets for downstream predictors and reinforcement-learning agents [24, 51], and the generate-then-optimize pipeline of §5 [5, 17] closes the loop by optimizing the allocation directly against generated scenarios. Across all three routes the strongest results couple a financial inductive bias to the generator rather than relying on a generic backbone. The recurring caveat is evaluation: portfolio metrics are especially vulnerable to look-ahead bias and to overfitting a single backtest, a theme we return to next.

8 EMERGING AND CROSS-DOMAIN APPLICATIONS

Beyond the five core tasks, several adjacent applications are maturing. *Fixed income* has its first generative treatments—arbitrage-aware term structures [21] and Treasury-futures synthesis [84]—extending diffusion from equities to rates. *Insurance and actuarial* use is nascent but methodologically close: diffusion for survival (time-to-event) data [10] maps onto mortality and claims modeling. *Macro-conditioned* generation links return distributions to economic state, a capability shared by factor-conditioned models [22] and macro-aware robust-trading generators [79]. Finally, much methodology transfers from *energy* markets, where conditional diffusion for scenario generation is comparatively mature [44, 73]; these electricity- and mobility-scenario techniques are routinely validated before adaptation to financial risk, and the multistage-stochastic-optimization framing of scenario trees [82] applies equally to energy arbitrage and to financial asset–liability management. The pattern is consistent: a technique proven on a structurally similar non-financial problem is imported and specialized with a financial prior.

9 CROSS-CUTTING THEMES

Read across tasks, four threads organize the field. **Conditioning is the norm, not the exception.** The most recent and most finance-specific work—controllable generation [70], counterfactual microstructure [76, 77], stress scenarios [5, 17], and forecast-conditioned allocation [16, 22]—is conditional, because controllability is what finance actually wants from a generator. **Inductive bias beats architecture tuning.** The wins come from encoding financial structure—GBM noising [37], factor covariance [13], arbitrage penalties [35], risk-neutral drift shifts [72], correlation regularizers [16]—into the noise process, the prior, or the loss. **Representation is a design lever.** Wavelet-to-image [69], LOB-as-image [6], byte-space [45], latent [57, 83], and chart-image [41] framings repeatedly unlock generic diffusion machinery for financial objects. **Generation increasingly serves a decision.** The frontier couples the generator to control—deep hedging [11, 70], CVaR allocation [5], multistage stochastic optimization [82], and the diffusion-as-policy line [2, 34]—rather than treating realistic samples as an end in themselves.

Deployment considerations. Three practical constraints cut across all of the above. *Latency*: real-time and high-frequency use cannot afford many-step sampling, so few-step methods [47, 64, 65] and skip-step schemes [86] are load-bearing, not optional. *Data governance*: when synthetic data substitutes for sensitive records, differential privacy and membership-inference resistance become hard requirements [60, 62, 88]. *Validation*: any generator feeding capital, valuation, or surveillance inherits model-risk-management duties—a generator used to set a capital number must itself be validated, monitored, and auditable.

10 EVALUATION, BENCHMARKS, REPRODUCIBILITY

Evaluation is the weakest link, and matters more in finance than in vision: a generator that looks good by a generic metric can misprice tail risk or leak look-ahead information. *Stylized-fact* fidelity (excess kurtosis, the Hill tail index, autocorrelation of returns vs. absolute returns) is the first-line test [37, 69], with optimal-transport distances a more interpretable, mechanism-attributing alternative for market simulators [26]; these checks are necessary but largely marginal. *Distributional/downstream* metrics probe joint and task-relevant structure: discriminative/predictive scores [9] and train-on-synthetic/test-on-real [58], the operational standard for replacing real data. Microstructure has LOB-Bench [6], though *responsiveness* (impact realism under the simulated agent's own

Method family	Synthetic	LOB	Risk	Pricing	Portfolio
DDPM / score-SDE	●	●	●	●	●
Conditional / guided	●	●	●	●	●
Structure-informed	●	●	●	●	●
Latent / tabular	●	●	●	●	●
Flow / interpolant	●	●	●	●	●

Figure 2: Qualitative coverage map of the surveyed literature. Larger/darker markers indicate denser activity in this survey, not comparable publication counts. The map highlights the current imbalance: conditioning is widespread, while flow/interpolant methods and hard-constrained pricing remain early.

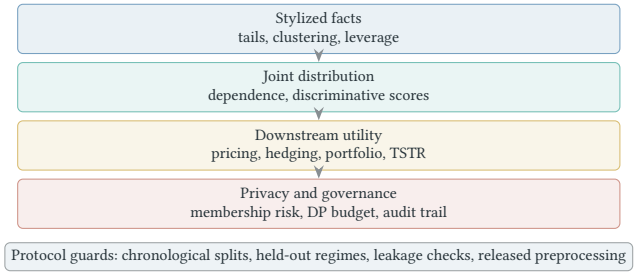


Figure 3: Evaluation stack for financial diffusion models. Each layer catches a different failure mode; credible studies should report at least one check at each layer rather than relying on marginal stylized facts alone.

actions) remains non-standardized. *Privacy* is first-class when synthetic data substitutes for sensitive records [59, 62, 88]. It is useful to view these as four metric families—marginal (stylized facts), joint/distributional (discriminative scores), task (downstream utility), and privacy—each guarding against a different failure: a model can ace marginals yet fail joint dependence, or attain high utility yet leak membership. A credible evaluation reports at least one from each family; reporting only marginals, still common, is the weakest acceptable bar and the easiest to game.

Three pitfalls undermine comparability. (i) *Heterogeneous data and metrics*: papers use different assets, periods, and scores, so reported superiority rarely transfers. (ii) *Leakage and look-ahead bias*: full-sample normalization or conditioning on information unavailable at decision time inflates results, especially in forecasting and portfolio studies. (iii) *In-distribution evaluation*: testing on the training regime hides the non-stationarity failure that matters in deployment. We recommend chronological splits, regime-shifted test sets, fixed public benchmarks, and released code; the absence of a shared, finance-specific benchmark suite is the single biggest obstacle to measurable progress.

A recommended reporting protocol. Concretely, we suggest each study report: (i) the exact data source, asset universe, and date range; (ii) a *chronological* train/test split with an explicit out-of-regime test period (e.g. a held-out crisis); (iii) at least one stylized-fact metric, one discriminative/predictive score, and one downstream-task (train-on-synthetic, test-on-real) metric, since each probes a different failure mode; (iv) for private data, a stated privacy notion and budget rather than an empirical proxy alone; and (v) released code and preprocessing. None of these is universal today, which is why headline claims of superiority so rarely reproduce across papers.

11 OPEN PROBLEMS

Tail and extreme-event fidelity. Decision-relevant mass lives in the tails, yet strong results degrade there [85], and risk numbers inherit any tail thinning. Objectives and noise schedules that explicitly target tail accuracy, and rare-event conditioning, are open; structure-informed designs [13, 37] help but do not solve it. **Non-stationarity.** Out-of-regime generalization and guarantees under shift are largely unaddressed; existing guarantees are narrow [13]. **Hard economic constraints.** Enforcing exact no-arbitrage and the $\mathbb{Q}$-martingale property by construction (not penalty [35]) is a prerequisite for front-office adoption. **Efficiency.** Few-step sampling [47, 64, 65] for real-time/HFT is underexplored on heavy-tailed targets and is essential for responsive simulation. **Decision integration.** Coupling generation to control [8, 17, 23] is promising but immature. **Governance.** Model risk, auditability, and formal privacy [62] gate regulated use. **Causality and counterfactuals.** Most models are correlational generators; counterfactual microstructure [76, 77] and controllable agents [32] gesture toward intervention, but principled causal diffusion—answering "what if this policy or shock occurred"—is largely unexplored and is precisely what stress testing requires. **Theory.** Approximation and sample-complexity guarantees remain narrow: factor-structure results [13] and the control-theoretic view [8] are starting points, but error bounds under non-stationarity and for heavy-tailed targets are open. **Benchmarks.** A shared finance-specific suite—returns, LOB [6], tabular [59], and risk—with chronological, regime-aware splits would do for this field what community benchmarks did for vision diffusion.

12 CONCLUSION

Diffusion models have earned a permanent place in the quantitative-finance toolkit, mapping their stable training, mode coverage, conditioning, and SDE structure onto finance's needs across data generation, microstructure simulation, risk, pricing, and portfolio decisions. The evidence is encouraging, but the field's progress is harder to measure than to produce: tail fidelity degrades where finance cares most, evaluation is fragmented and frequently in-distribution, economic constraints are penalized rather than enforced, sampling is too slow for real-time use, and governance for regulated deployment is barely begun. The next phase depends less on new architectures than on discipline: finance-specific benchmarks with

chronological and regime-aware splits, tail- and arbitrage-aware objectives, formal privacy and model-risk treatment, and tighter coupling of generation to the decision it serves.

Scope and limitations of this survey. We have emphasized breadth across financial tasks over exhaustive coverage of any one; some fast-moving sub-areas—multimodal (news/price) generation, graph-based systemic-risk modeling, and crypto-specific work—are touched only briefly and merit dedicated treatment. Because the literature is recent and largely preprint-stage, reported results should be read as indicative rather than settled, which reinforces our call for shared benchmarks. We expect the method × task map of Figures 1–2 to fill in rapidly, particularly in the under-served cells of hard-constrained pricing and tail-faithful risk.

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